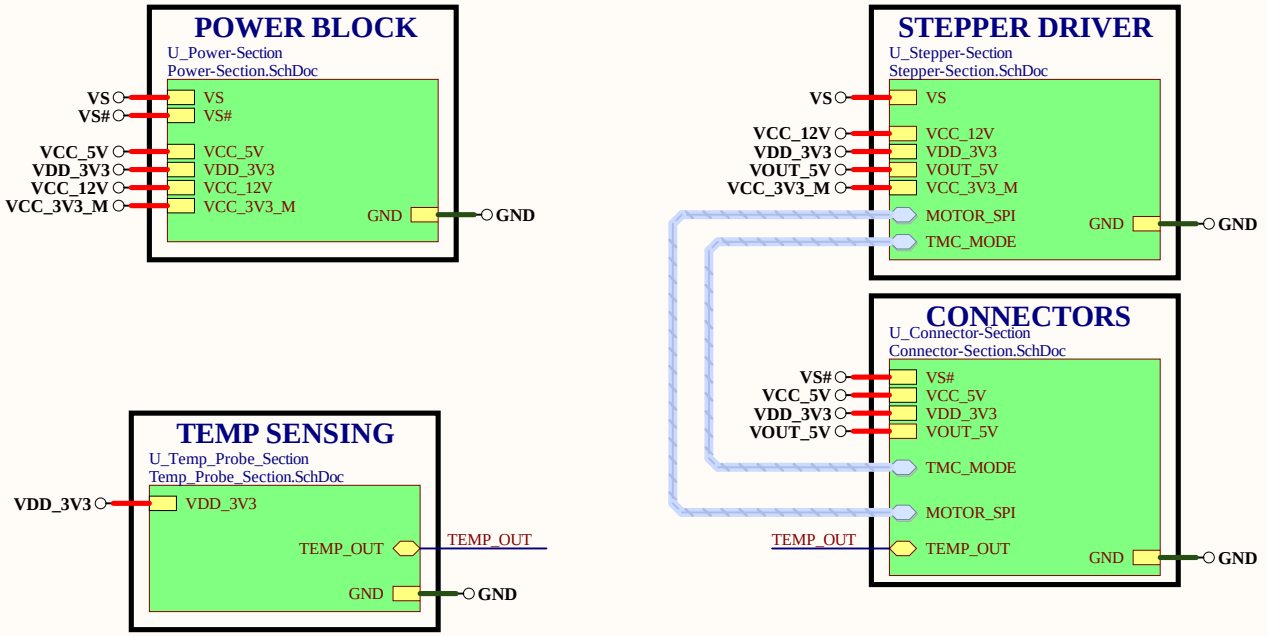
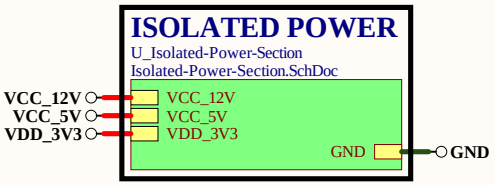
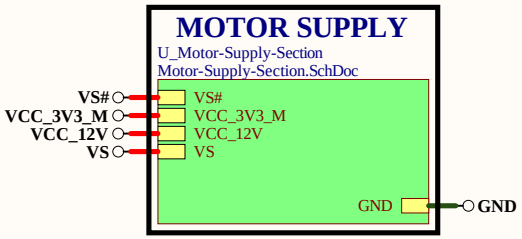
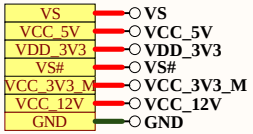
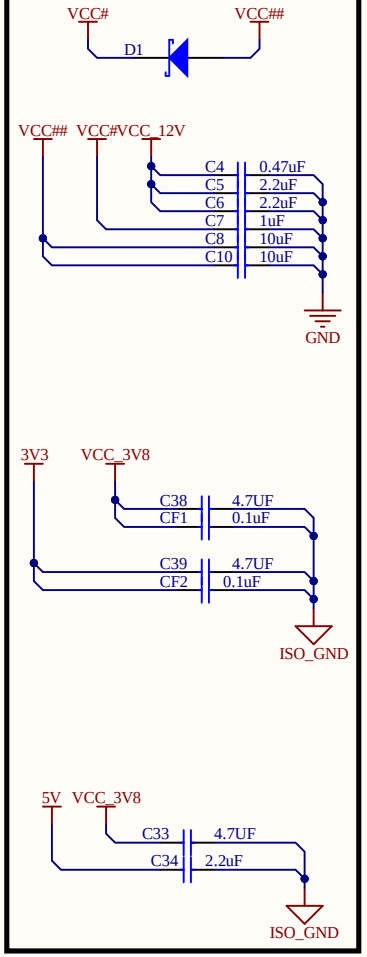

POWER BLOCK
U_Power-Section
Power-Section.SchDoc
VS
VS#
VCC_5V
VDD_3V3
VCC_12V
VCC_3V3_M
GND
GND
STEPPER DRIVER
U_Stepper-Section
Stepper-Section.SchDoc
VS
VCC_12V
VDD_3V3
VOUT_5V
VCC_3V3_M
MOTOR_SPI
TMC_MODE
GND
GND
CONNECTORS
U_Connector-Section
Connector-Section.SchDoc
VS#
VCC_5V
VDD_3V3
VOUT_5V
TMC_MODE
MOTOR_SPI
TEMP_OUT
GND
GND
TEMP SENSING
U_Temp_Probe_Section
Temp_Probe_Section.SchDoc
VDD_3V3
TEMP_OUT
GND
GND


ISOLATED POWER
U_Isolated-Power-Section
Isolated-Power-Section.SchDoc
VCC_12V
VCC_5V
VDD_3V3
GND
VCC_12V
VCC_5V
VDD_3V3
GND
GND

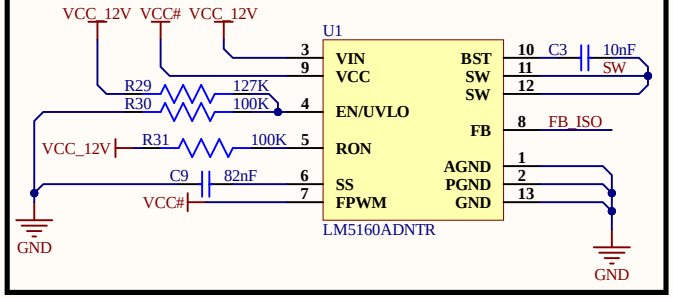

MOTOR SUPPLY
U_Motor-Supply-Section
Motor-Supply-Section.SchDoc
VS#
VCC_3V3_M
VCC_12V
VS
VS#
VCC_3V3_M
VCC_12V
VS
GND
GND


VS
VCC_5V
VDD_3V3
VS#
VCC_3V3_M
VCC_12V
GND
VS
VCC_5V
VDD_3V3
VS#
VCC_3V3_M
VCC_12V
GND

DECOUPLING CAPS

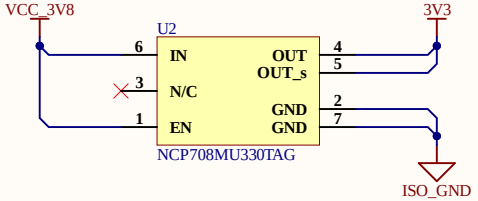


12V-3.8V ISOLATED BUCK CONVERTER

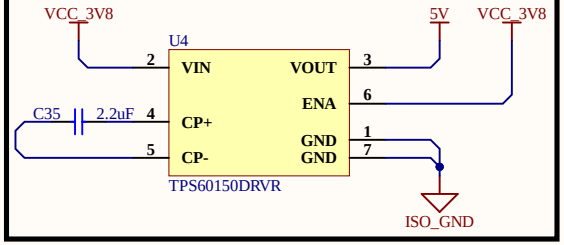


Primary side output voltage would be 5.7 for a 3.8V VSec (NP/Ns=Vpri/Vsec)
FB resistors RF2=3.7K and RF1=2K.
Max current of this transformer is 840mA. (suggested in datasheet)

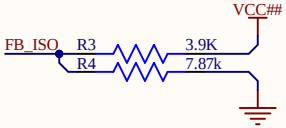
3.8-3.3V LINEAR REGULATOR



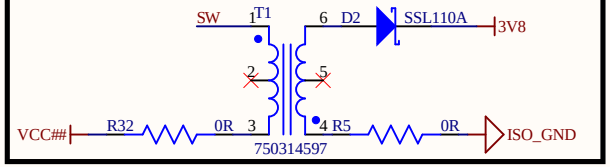
3.8-5V CHARGE PUMP



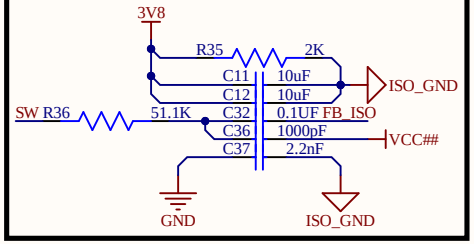
FEEDBACK NETWORK



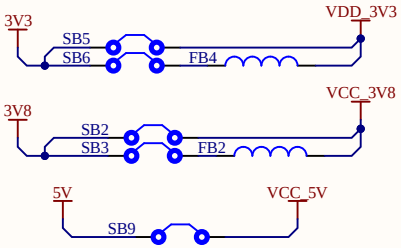
FLYBACK TRANSFORMER

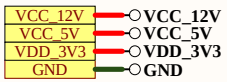


SUPPORTING CIRCUIT

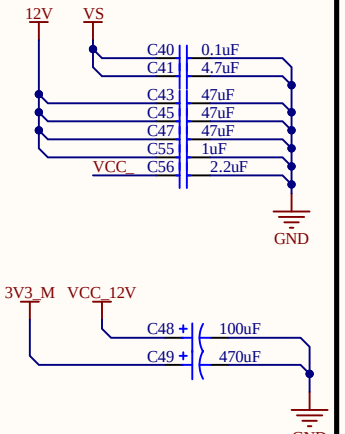


SOLDER BRIDGE CONNECTIONS

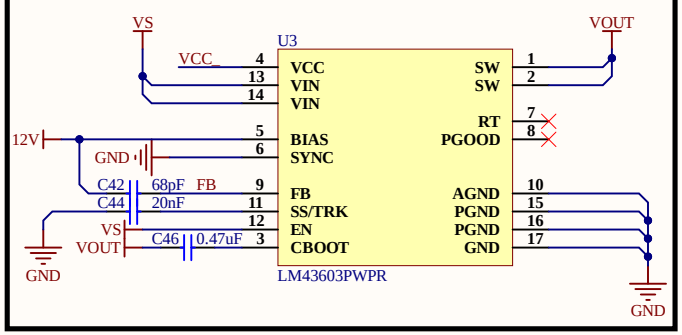




DECOUPLING CAPS



30V-12V NON-ISOLATED BUCK CONVERTER

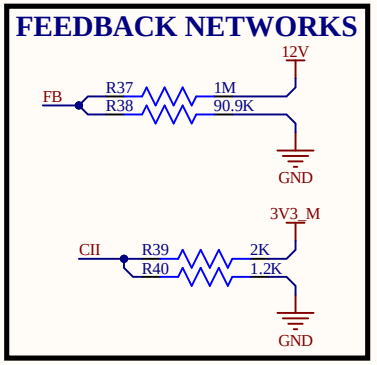


For VOUT=12V, RFBB= 90.9K, RFBT=1M (Datasheet pg:26)

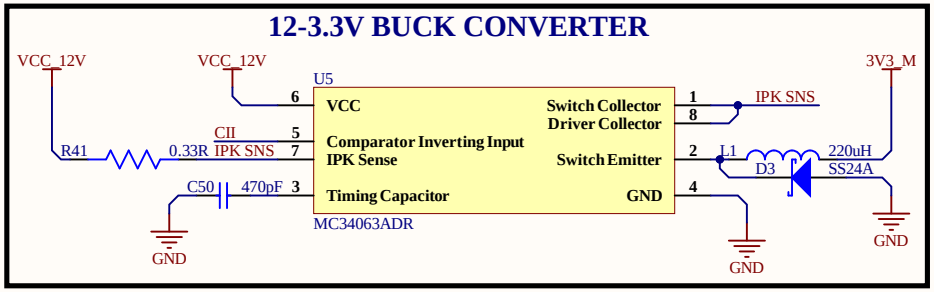
INDUCTOR



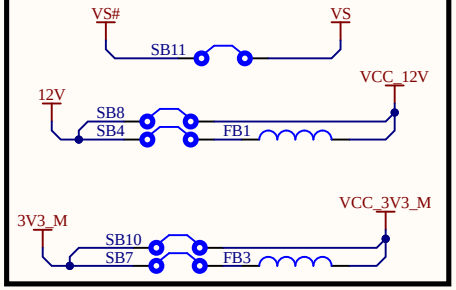
FEEDBACK NETWORKS

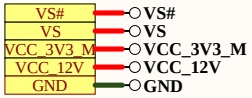


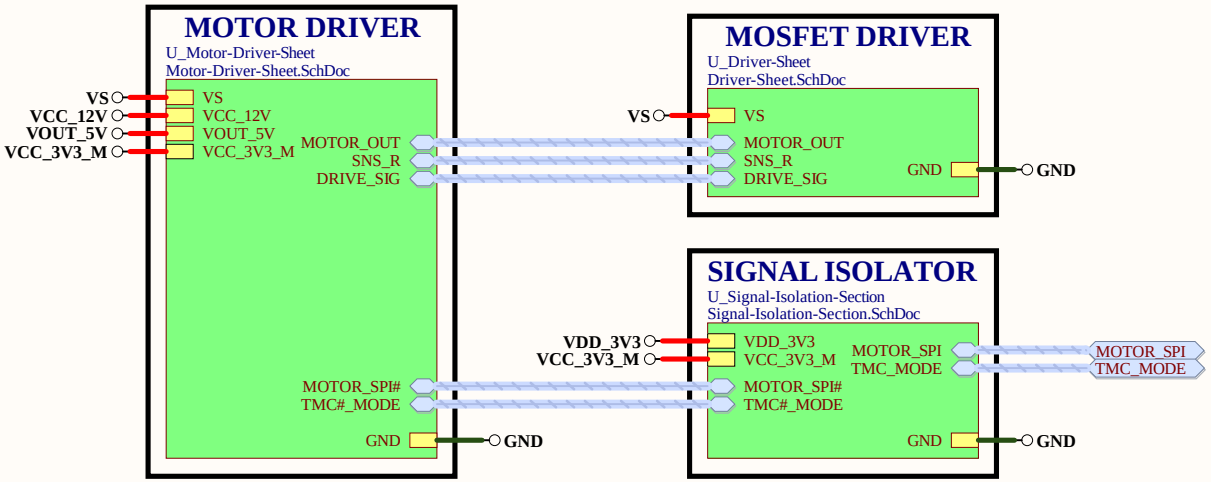
12-3.3V BUCK CONVERTER

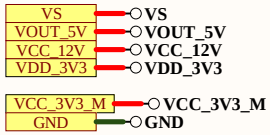


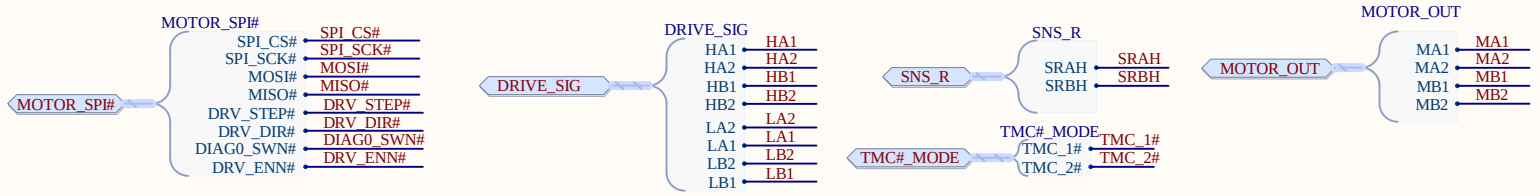
SOLDER BRIDGE



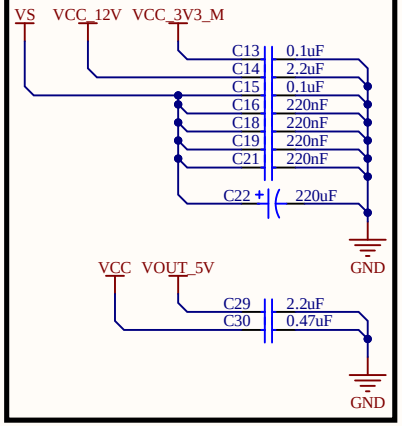



MOTOR DRIVER
U_Motor-Driver-Sheet
Motor-Driver-Sheet.SchDoc
VS
VCC_12V
VOUT_5V
VCC_3V3_M
VS
VCC_12V
VOUT_5V
VCC_3V3_M
MOTOR_OUT
SNS_R
DRIVE_SIG
MOTOR_SPI#
TMC#_MODE
GND
GND
MOSFET DRIVER
U_Driver-Sheet
Driver-Sheet.SchDoc
VS
VS
MOTOR_OUT
SNS_R
DRIVE_SIG
GND
GND
SIGNAL ISOLATOR
U_Signal-Isolation-Section
Signal-Isolation-Section.SchDoc
VDD_3V3
VCC_3V3_M
VDD_3V3
VCC_3V3_M
MOTOR_SPI
TMC_MODE
MOTOR_SPI#
TMC#_MODE
GND
GND
MOTOR_SPI
TMC_MODE

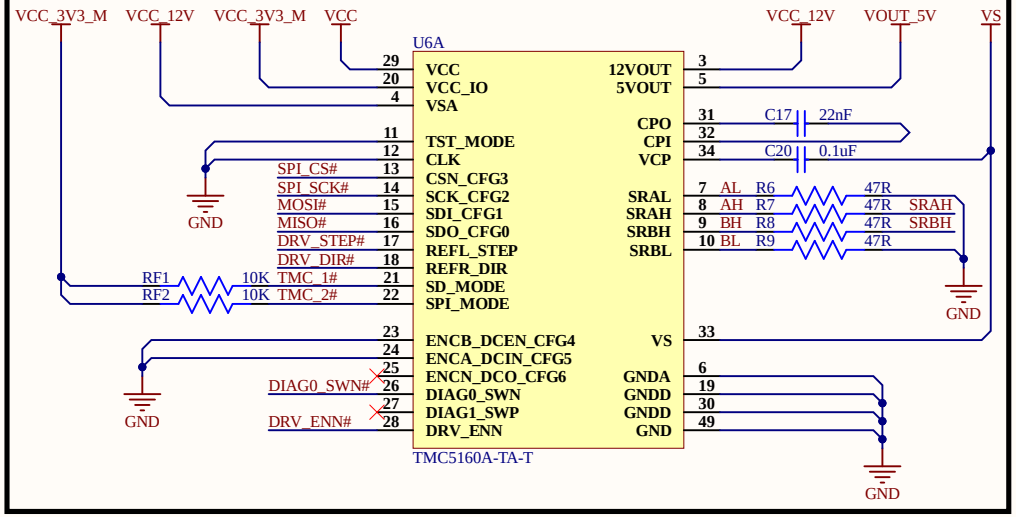

VS
VOUT_5V
VCC_12V
VDD_3V3
VS
VOUT_5V
VCC_12V
VDD_3V3
VCC_3V3_M
GND
VCC_3V3_M
GND



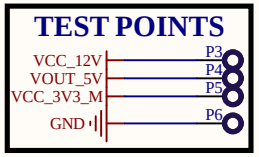
DECOUPLING CAPS



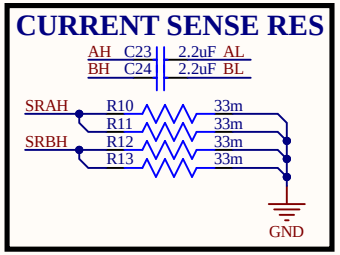
STEPPER MOTOR DRIVER-PART-A



TEST POINTS



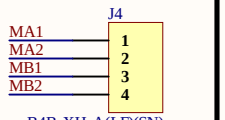
CURRENT SENSE RES



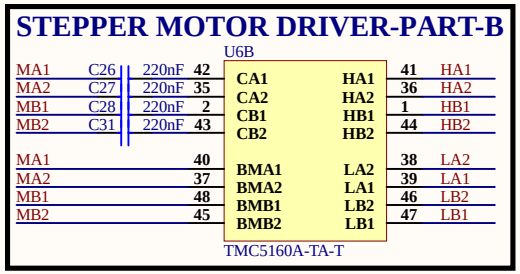
SERIES RESISTOR

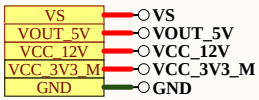


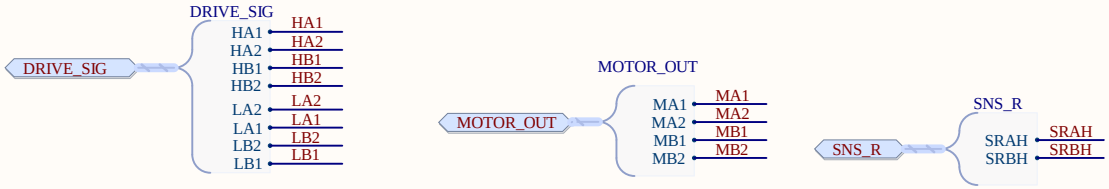
MOTOR OUTPUT



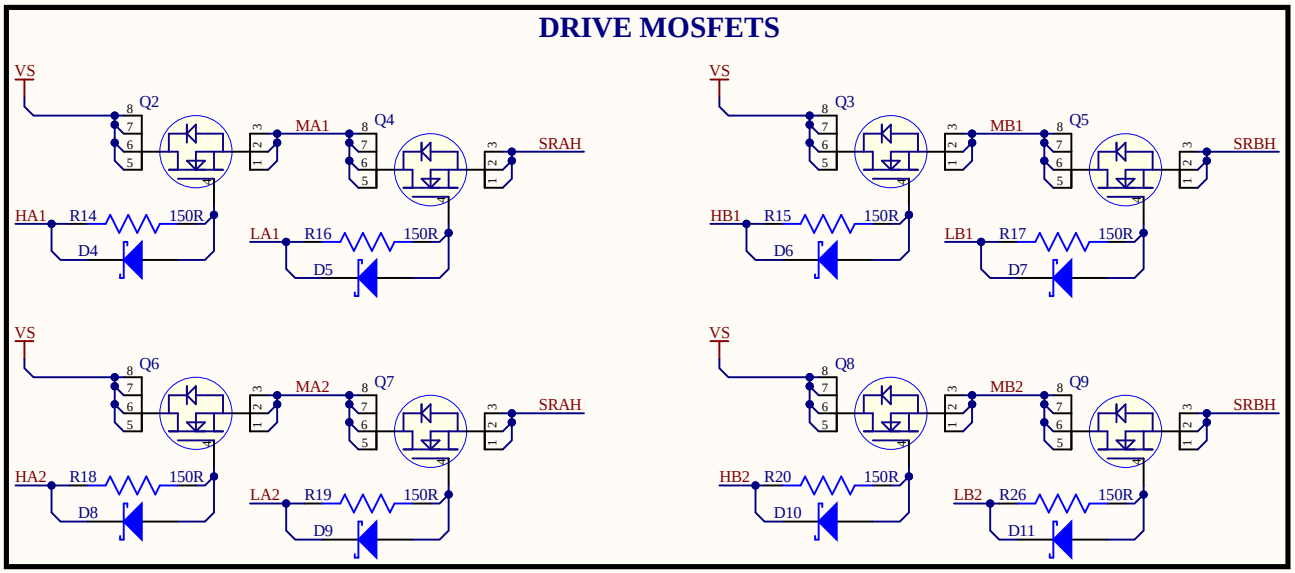
STEPPER MOTOR DRIVER-PART-B



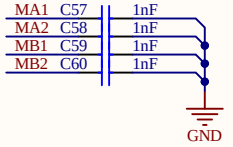




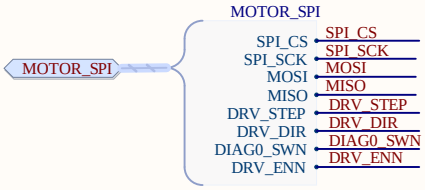
DRIVE MOSFETS

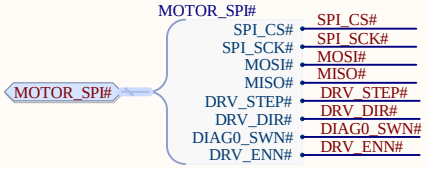


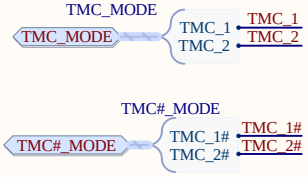
DECOUPLING CAPS

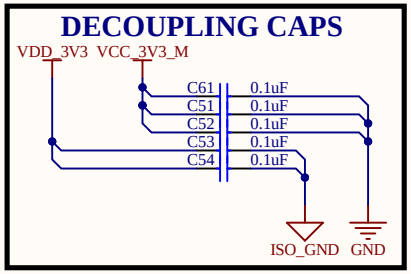


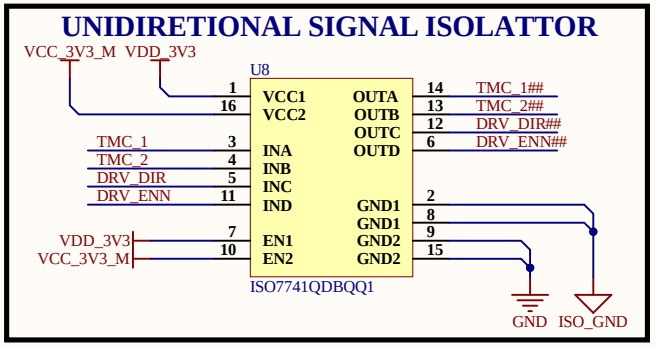


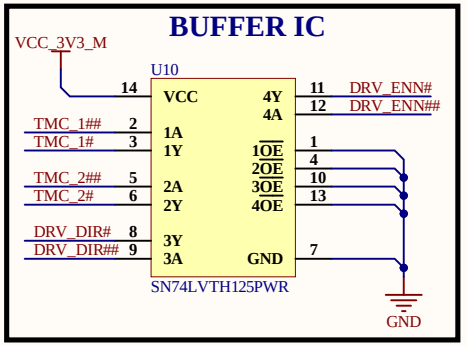


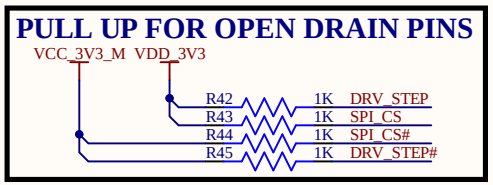


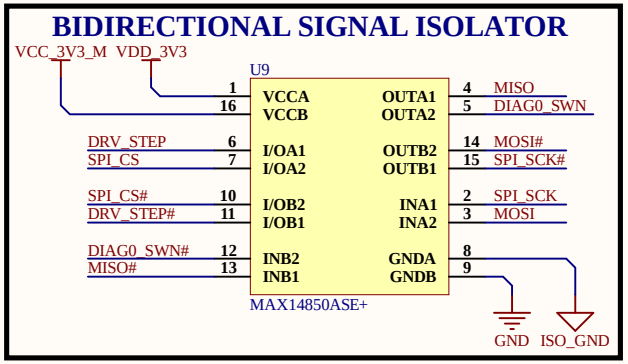










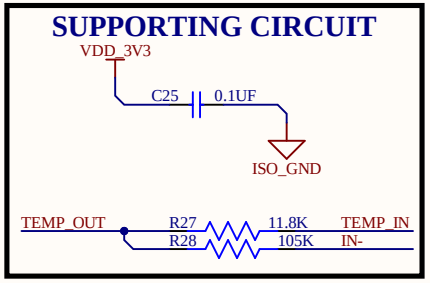




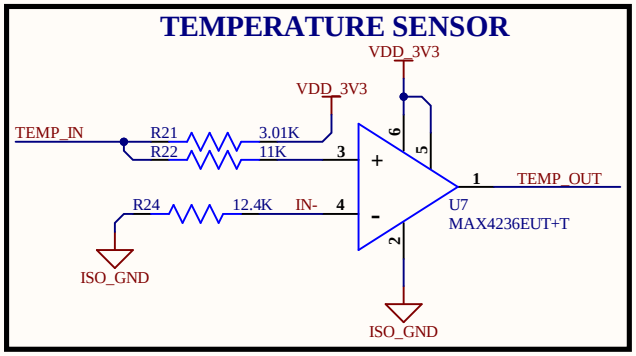
Title		
Size A4	Number	Revision
Date: 8/05/2022		Sheet of
File: E:\ColeBenson-SmartMotors\..\Signal-Isolation-Section.SchDoc		Drawn By:

TEMP_OUT TEMP_OUT

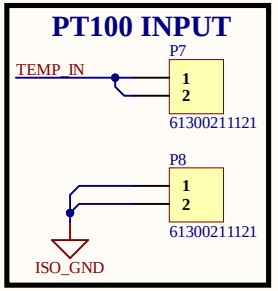
SUPPORTING CIRCUIT



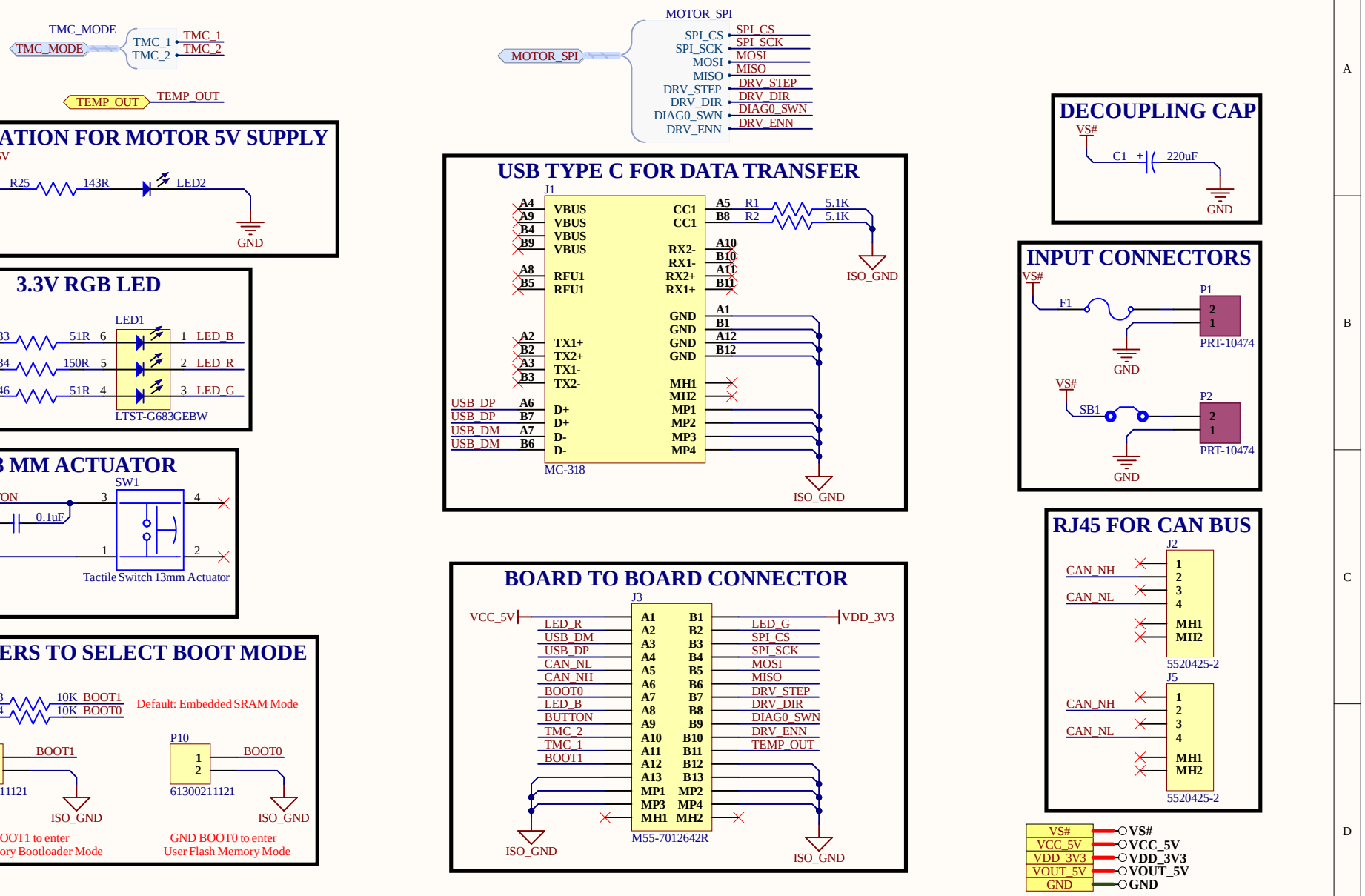
TEMPERATURE SENSOR

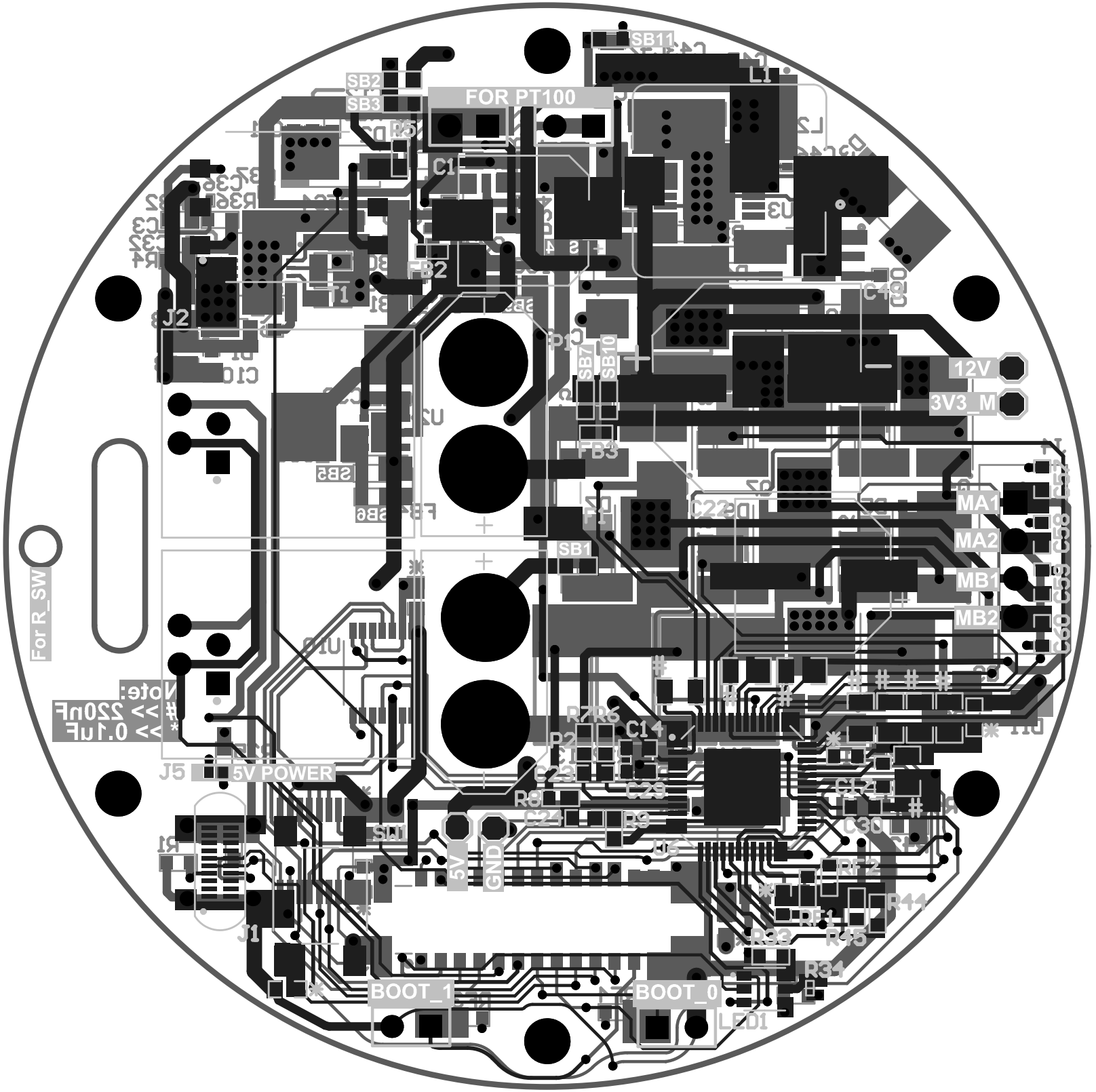


PT100 INPUT



VDD_3V3 **VDD_3V3**
GND **GND**


TMC_MODE
TMC_MODE
TMC_1
TMC_2
TMC_1
TMC_2
TEMP_OUT
TEMP_OUT
MOTOR_SPI
MOTOR_SPI
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
INDICATION FOR MOTOR 5V SUPPLY
VOUT_5V
R25
143R
LED2
GND
3.3V RGB LED
VDD_3V3
LED1
R33
51R
6
1
LED_B
R34
150R
5
2
LED_R
R46
51R
4
3
LED_G
LTST-G683GEBW
13 MM ACTUATOR
SW1
BUTTON
3
4
C2
0.1uF
1
2
Tactile Switch 13mm Actuator
ISO_GND
HEADERS TO SELECT BOOT MODE
VDD_3V3
RF3
10K
BOOT1
RF4
10K
BOOT0
Default: Embedded SRAM Mode
P9
1
2
BOOT1
61300211121
ISO_GND
GND BOOT1 to enter
System Memory Bootloader Mode
P10
1
2
BOOT0
61300211121
ISO_GND
GND BOOT0 to enter
User Flash Memory Mode
USB TYPE C FOR DATA TRANSFER
J1
A4
A9
B4
B9
VBUS
VBUS
VBUS
VBUS
A8
B5
RFU1
RFU1
A2
B2
A3
B3
TX1+
TX2+
TX1-
TX2-
USB_DP
A6
USB_DP
B7
USB_DM
A7
USB_DM
B6
D+
D+
D-
D-
CC1
CC1
RX2-
RX1-
RX2+
RX1+
GND
GND
GND
GND
MH1
MH2
MP1
MP2
MP3
MP4
A5
B8
R1
R2
5.1K
5.1K
ISO_GND
A10
B10
A11
B11
A1
B1
A12
B12
ISO_GND
MC-318
BOARD TO BOARD CONNECTOR
J3
VCC_5V
LED_R
USB_DM
USB_DP
CAN_NL
CAN_NH
BOOT0
LED_B
BUTTON
TMC_2
TMC_1
BOOT1
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
A13
MP1
MP3
MH1
B1
B2
B3
B4
B5
B6
B7
B8
B9
B10
B11
B12
B13
MP2
MP4
MH2
LED_G
SPI_CS
SPI_SCK
MOSI
MISO
DRV_STEP
DRV_DIR
DIAG0_SWN
DRV_ENN
TEMP_OUT
VDD_3V3
ISO_GND
M55-7012642R
ISO_GND
DECOUPLING CAP
VS#
C1
220uF
GND
INPUT CONNECTORS
VS#
F1
P1
2
1
PRT-10474
GND
VS#
SB1
P2
2
1
PRT-10474
GND
RJ45 FOR CAN BUS
J2
CAN_NH
CAN_NL
1
2
3
4
MH1
MH2
5520425-2
J5
CAN_NH
CAN_NL
1
2
3
4
MH1
MH2
5520425-2
VS#
VCC_5V
VDD_3V3
VOUT_5V
GND
VS#
VCC_5V
VDD_3V3
VOUT_5V
GND


SB11
SB2
SB3
FOR PT100
R5
C1
FB2
J2
C10
C40
SB7
SB10
12V
3V3_M
FB3
SB5
SB6
C22
MA1
MA2
MB1
MB2
SB1
For R_SW
Note:
>> 220nF
* >> 0.1uF
R7
R6
C14
P2
C23
C29
R8
C24
R9
J5
5V POWER
SW
5V
GND
C30
RF2
R41
RF1
R39
R45
R34
J1
BOOT_1
BOOT_0
LED1